their direction of motion—and these changes can be felt. For instance, if George were wearing a jet-pack firing away from his back, he would definitely feel that he was moving. This feeling is intrinsic. If the jet-pack is firing away, George knows he is moving, even if his eyes are closed and therefore can make no comparisons with other objects. Even without such comparisons, he would no longer claim that he was stationary while “the rest of the world was moving by him.” Constant-velocity motion is relative; not so for non-constant-velocity motion, or, equivalently, accelerated motion. (We will re-examine this statement in the next chapter when we take up accelerated motion and discuss Einstein’s general theory of relativity.)

Setting these stories in the darkness of empty space aids understanding by removing such familiar things as streets and buildings, which we typically, although unjustifiably, accord the special status of “stationary.” Nonetheless, the same principle applies to terrestrial settings, and in fact is commonly experienced.1 For example, imagine that after you have fallen asleep on a train, you awake just as your train is passing another on adjacent parallel tracks. With your view through the window completely blocked by the other train, thereby preventing you from seeing any other objects, you may temporarily be uncertain as to whether your train is moving, the other train is moving, or both. Of course, if your train shakes or jostles, or if the train changes direction by rounding a bend, you can feel that you are moving. But if the ride is perfectly smooth—if the train’s velocity remains constant—you will observe relative motion between the trains without being able to tell for certain which is moving.

Let’s take this one step further. Imagine you are on such a train and that you pull down the shades so that the windows are fully covered. Without the ability to see anything outside your own compartment, and assuming that the train moves at absolutely constant velocity, there will be no way for you to determine your state of motion. The compartment around you will look precisely the same regardless of whether the train is sitting still on the tracks or moving at high speed. Einstein formalized this idea, one that actually goes back to insights of Galileo, by proclaiming that it is impossible for you or any fellow traveler to perform an experiment within the closed compartment that will determine whether or not the train is moving. This again captures